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A Century of Vegetation Change on an Offshore Islet¹

Brandon P. C. Najarian,^{2,3} Kevin Faccenda,² and Kasey E. Barton²

Abstract: Offshore islets are features of archipelagos and coastal continental regions. Because of their small size, they can be considered microcosms for studies of island biogeography and community assembly. The Hawaiian Islands include at least 54 islets, which provide safe habitat for nesting seabirds and monk seals due to the absence of mammalian predators and laws restricting human activity. Because the Hawaiian islets are uninhabited by people, they offer unique opportunities to examine community dynamics in relatively undisturbed habitats. Mānana Islet, also known as Rabbit Island, is a 25.5-hectare islet located off Kaupō Beach in Waimānalo, O‘ahu Island, that has been designated a Seabird Sanctuary by the state of Hawai‘i. European rabbits (*Oryctolagus cuniculus*) were introduced to Mānana in the 1880s for food and game but were eradicated by 1985 through a state-initiated hunting and poisoning program. Across a nearly 80-yr period (1927–2008), spanning rabbit habitation and eradication, the vegetation of Mānana has been surveyed or reported 21 times, providing a longitudinal record of plant diversity on the islet. To extend this record, a new vegetation survey was conducted in 2023, replicating previous methods. Rabbits had devastating effects on the vegetation of Mānana, which then increased in plant species richness after rabbit eradication. The most recent surveys have detected an increase in non-native species, highlighting biological invasion as an ongoing threat to these islet habitats. In the most recent survey, a total of 43 plant species were detected, including seven new records, six of which are non-native. How these patterns in species richness correspond to measures of diversity based on abundance and evenness remains unclear. Additional restoration of Mānana Islet may increase native biodiversity, supporting nesting seabirds and ensuring these diverse coastal islets persist in the future.

Keywords: Mānana, islets, Hawai‘i, island biodiversity, survey, vegetation, plant invasion

THE STUDY OF ISLAND biogeography has a long history, producing foundational theories for ecology and evolution (MacArthur and Wilson 1967). More recently, island biodiversity

has been in tremendous flux as native species decline under global change threats and non-native species spread (Russell and Kueffer 2019). Because islands are home to a disproportionate extent of global biodiversity due to high rates of endemism (Fernández-Palacios et al. 2021), documenting and mitigating this turnover in community assembly are major conservation priorities. Offshore islets, generally defined as small land masses unable to sustain human communities, are important components of archipelagos, although they rarely feature in studies of island biodiversity. Yet, because of their small size, they offer tractable sites (i.e., microcosms) for characterizing patterns of diversity. The small island effect suggests that very small islands,

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FIGURE 1. Aerial photo of Mānana Islet, located off Kaupō Beach, Waimānalo, Hawaiʻi. Image provided by Jorelle Yuvienco.

such as islets, are constrained by their limited area and elevation, supporting relatively few species with small population sizes (Rand and Rabor 1960). At the extreme, the relationship between island area and species richness is believed to become decoupled, meaning that changes in area no longer significantly influence the number of species present (Lomolino and Weiser 2001). Species turnover across time is also expected to be high in nearshore islets as a consequence of high dispersal rates to islets and high extinction rates associated with the small population sizes (MacArthur and Wilson 1967).

Species turnover has been documented to be very high on islets of the Aegean Sea (Panitsa et al. 2008) and on lake islands in Sweden (Nilsson and Nilsson 1982), illustrating the highly dynamic community assembly of islet floras. These dynamic communities reflect fluctuations of small populations in response to natural disturbances (Simberloff and Wilson 1969, Vidal et al. 2000), but anthropogenic activities are introducing novel disturbances that may exacerbate the naturally dynamic islet communities. The strong effects of human activities on island community assembly are well documented globally (Nogué et al. 2021), resulting primarily from human alterations to the landscape (clearing for human settlements and cultivation) and introductions of non-native species with often devastating effects on the native island species. While most studies have focused on

anthropogenic effects on larger, main island communities, we predict that these effects may be even stronger on islet communities given the reduced area and small population sizes typical of species on islets.

The Hawaiian Archipelago offers compelling opportunities to investigate plant community dynamics in offshore islets. Located over 3,000 km from any continental land mass, Hawaiʻi is the most isolated archipelago on the planet, and the flora has evolved *in situ* from a mere 280 colonizing plants into approximately 1,400 angiosperm species, with more than 90% endemism (Wagner et al. 2005, Price and Wagner 2018). The Hawaiian Archipelago consists of 8 main islands as well as 54 smaller offshore islets (Offshore Islet Project 2008). These islets range dramatically in size, from less than 0.4 to over 109 hectares, and collectively comprise 303 hectares of land. They serve as nesting sites for hundreds of thousands of seabirds and many endemic plant species (Offshore Islet Project 2008). The islets also vary widely in age, with some forming just 50,000 to 70,000 years ago during the rejuvenation stage of volcanism. In contrast, other islets are millions of years old and were once part of a main island before the collapse of a land bridge left them isolated. This diversity in formation is reflected in the islets' composition and substrate—those formed more recently during submarine rejuvenation eruptions are primarily composed of palagonite, while many of the older islets are basaltic (Wentworth 1926). Because of their extremely small size and saline influence, the Hawaiian islets are composed entirely of coastal habitat and associated species.

Across the Hawaiian Archipelago (including islets and main islands), coastal ecosystems represent a significant portion of the total land cover and provide ecological and cultural services that include recreation, carbon sequestration, and erosion reduction (UNEP 2006, Duarte et al. 2013, Farrer et al. 2022). Hawaiʻi has approximately 750 miles of coastline, making it the fourth longest in the United States (Beaver 2006). Within coastal habitats, there are many indigenous species (defined as those native to Hawaiʻi and elsewhere),

making the endemism rate lower than that of other habitat types (Richmond and Mueller-Dombois 1972). Yet, most coastal plant species are restricted to these narrow habitats and so represent a unique component of island biodiversity (Warshauer et al. 2009). Hawai'i's coastal plants also foster associated biodiversity, providing critical habitat for nesting seabirds and breeding marine mammals, as well as floral resources for endemic bees (Duffy 2010, Croxall et al. 2012, Magnacca 2020). Nesting seabirds are particularly abundant in Hawai'i's coasts and islets, with nearly 15 million birds from 22 species making yearly nests on the islands (Harrison 1990, Duffy 2010), surpassing other states with extensive coastlines, such as California, which has only 2 million birds nesting in coastal habitats each year (Carter et al. 1995). Islets play a crucial role in seabird conservation, as they provide a safer habitat with fewer introduced predators compared to the main Hawaiian Islands. In addition, native vegetation contributes to suitable nesting structures and supports essential breeding activities (Troy et al. 2017).

Globally, coastal ecosystems are under severe threat from commercial development, climate change, and the establishment of invasive plant and animal species (Nogué et al. 2021, Calvin et al. 2023, Steven et al. 2023). In coastal ecosystems, invasive species can take over large areas of coastal habitats and facilitate the establishment of other non-native invasive species (Demopoulos and Smith 2010, Kueffer et al. 2010). Invasive species are also a major threat to island biodiversity (Fernández-Palacios et al. 2021), and in Hawai'i, invasive species effects are magnified due to limited land availability, small native plant population sizes, and extreme isolation (Daehler et al. 2004, Brock and Daehler 2021). Despite the relatively small size of Hawai'i, 44% of all the threatened and endangered species in the United States occur here (Rare Plant Program 2013). Identifying incipient invasions through early detection and rapid response through repeated surveys offer one of the most effective strategies for limiting the spread of non-native species (Reaser et al. 2020, Yemshanov et al. 2022). Due to their small size, offshore islets are ideal targets to monitor

for invasive plants regularly to conserve these ecologically important habitats.

In addition to invasive plants, invasive animals are major threats to island biodiversity (Caujapé-Castells et al. 2010). Non-native mammals, particularly rabbits, cats, rats, goats, and pigs, are hugely problematic on islands, directly and indirectly damaging native plants through consumption and habitat destruction (Courchamp et al. 2003). Rabbit species are now widespread, having been introduced to over 800 islands worldwide (Flux and Fullagar 1992). Through their selective feeding and propensity to establish large populations, rabbits have devastating effects on the vegetation in areas where they are introduced, leading to erosion and rapid declines in native plant communities, and there have been many efforts to control and eradicate rabbits from islands (Courchamp et al. 2003, Chapuis et al. 2004). Following rabbit eradication, native plants commonly increase in abundance and richness on islands and islets (North et al. 1994, Chapuis et al. 2004, Rita et al. 2022). Unfortunately, it has also been observed that non-native plants may increase in abundance and richness following rabbit eradication from islands (Chapuis et al. 2004, Eijzena 2011), revealing the complex and interactive dynamics of multiple threats to island biodiversity. Most of these studies have provided only short-term snapshots of vegetation recovery within a few years following rabbit eradication, and so the balance between native plant recovery and non-native plant invasion over longer time periods remains unclear.

In this study, we provide a longitudinal record of plant species diversity for an offshore islet in the Hawaiian Islands, Mānana Islet (Figure 1), on which rabbits were eradicated by 1985, nearly 40 yr ago (Swenson 1986). Combining recent fieldwork with historical records, we document 100 yr of vegetation dynamics spanning peak rabbit abundance, eradication, and subsequent recovery periods. Consistent with trends seen globally on oceanic islands (Kueffer et al. 2010), we predict that there has been a general increase in non-native species and a decrease in natives over 96 yr. However, we also predict that increases in both native and non-native species

richness will be detectable following rabbit eradication, as has been shown on other oceanic islets (Chapuis et al. 2004, Eijzena 2011). Our most recent survey in 2023 represents nearly 40 yr of rabbit-free vegetation dynamics, and so provides the opportunity to assess longer term recovery patterns than has been previously reported on islets.

MATERIALS AND METHODS

Study System

Mānana Islet is an uninhabited 25 -hectare islet located 1.2 km off Kaupō beach on the island of Oʻahu, Hawaiʻi (21° 19' 41" N, 157° 39' 25" W), and is the largest islet of Oʻahu. Mānana is composed of two tuff crater vents on the east and northwest sides (Wentworth 1926, Tomich et al. 1968). The islet formed during the secondary phase of Oʻahu eruptions over 40,000 years ago, with the summit currently reaching 110 m on the east vent, which has since eroded due to marine abrasion, resulting in a coastal bench measuring approximately 122 m wide and 305 m long with splash pools (Wentworth 1926, Bryan 1934). The northwest vent is mainly intact, with a crater floor dominated by non-native grasses, including *Cenchrus ciliaris*.

The State Seabird Sanctuary designation of Mānana creates a unique habitat that protects seabirds from many anthropogenic and mammalian threats, creating a safe nesting site unavailable on the main Hawaiian Islands (Harrison 1990), and attracting over 100,000 seabirds annually of differing species. The absence of mammalian predators is of particular significance because many of the seabird species that frequent the islet are ground-nesting seabirds, relying only on the vegetation for protection from predation (Richardson and Fisher 1950, Harrison 1990). Because of these unique ecological factors and large seabird colonies, the vegetation of Mānana has been surveyed 21 times over the past 100 yr, with the most recent comprehensive survey being completed in 2005 by Kim and Forest Starr (Starr and Starr 2006).

Mānana has cultural significance to Native Hawaiians who built two fishing shrines near the islet dedicated to ʻāholehole (*Kuhlia*

marginata) and moi (*Polydactylus sexfilis*) fish (Sterling and Summers 1993). These species were significant to Native Hawaiians because they were staples in their diet, and moi also represented a disastrous omen for chiefs (Andrews 1974). Records indicate that burials took place on the islet during ancient times (Sterling and Summers 1983). During the late 1800s, European rabbits (*Oryctolagus cuniculus*) were introduced to Mānana by Waimānalo rancher John Adams Kuakini Cummins for food and game hunting (Bryan 1915, Tomich et al. 1968, Clark 2002). Because there are no vegetation surveys prior to rabbit introduction, we lack information about the species present and their abundances before this disturbance (Tomich et al. 1968, Swenson 1986). Rabbits were allowed to roam and reproduce freely until 1962, when the State of Hawaiʻi authorized a hunting and poisoning program to eradicate the rabbits from Mānana (Tomich 1965). The program concluded in 1964 due to declining interest from hunters. In January 1965, zoologist Quentin Tomich reported in the Hawaiʻi Audubon Society journal *ʻElepaio* that the rabbit population did not negatively affect seabird nesting and described the relationship between the two species as harmonious. He even suggested that rabbits should be recognized as part of the Hawaiian fauna, having gained significance from a scientific perspective (Tomich 1965). Despite the removal efforts, seabird surveys in 1983 detected rabbits and their droppings on the islet, albeit in much lower numbers than before (Swenson 1986). In the particularly dry year of 1984, surveyors noted only one rabbit on the islet. By 1985, no rabbits or evidence of their presence were observed on Mānana, even after significant plant growth following heavy rains in May (Swenson 1986). Though these survey notes don't provide a definitive date of rabbit eradication on the islet, Swenson, who frequented the islet, notes that "If rabbits were still surviving on Manana in 1985, it seems likely that they would have been evident during these months of abundant green vegetation" (Swenson 1986).

The islet also played a role in the counter-revolution of 1895, when Royalist forces conducted military operations to overthrow

the Republic of Hawai‘i and return Queen Lili‘uokalani to her throne. On 30 December 1894, the American schooner H. C. *Wahlberg* landed on the islet and buried a cache of 285 guns and ammunition on the western shore. This cache was then retrieved by Royalist forces and utilized during the counter-revolution of 1895 (Office of the Historian 1895, Landgraf 1994). During WWII, the U.S. military also used Mānana as an artillery live fire training target (Green 1942). Even today, remnants of these training programs can be found on the islet as ammunition casings and spent bullets.

Historical Vegetation Surveys

From 1927 to 2008, 21 vegetation surveys were conducted on Mānana Islet (Table 1).

Many of these surveys were described in the Bishop Museum Offshore Project (Offshore Islet Project 2008). Archival searches were also performed by searching EBSCOhost and Web of Science using the search query “Mānana Islet OR Island,” “Rabbit Islet OR Island,” and “Kaohikaipu.” In addition, the *‘Elepaio* provided additional information on the history and vegetation of Mānana. Many of the historical surveys used a haphazard method due to rugged terrain and often only passively mentioned vegetation as it related to seabird nesting habitats. The spatial extent of the historical surveys also varied. Of the 20 historical surveys, eight were described as partially comprehensive, five were fully comprehensive, and seven did not mention the area covered during the surveys. Six surveys were personal communications,

TABLE 1
Compiled Vegetation Surveys Conducted on Mānana Islet during 100 Yr of Observation

Survey Date(s)	Survey Type	Observer	Source
1927	P	H. A. Lee	Tomich et al. (1968)
1930	<i>C_i</i>	<i>M. C. Neal</i>	<i>Bryan (1934), Tomich et al. (1968)</i>
1934	P _i	M. C. Neal	Bryan (1934), Tomich et al. (1968)
August 1934	P	E. H. Bryan Jr.	Bryan (1934)
1936	<i>C_i</i>	<i>F. R. Fosberg</i>	<i>Tomich et al. (1968)</i>
1937	P _i	F. A. Egler	Tomich et al. (1968)
1937–1941	P	F. Richardson and H. I. Fisher	Richardson and Fisher (1950)
1955	<i>C_i</i>	<i>C. H. Lamoureux</i>	<i>Tomich et al. (1968)</i>
February–March 1964	<i>C</i>	<i>Q. Tomich</i>	<i>Tomich et al. (1968)</i>
1964	<i>C</i>	<i>C. H. Lamoureux</i>	<i>Tomich et al. (1968)</i>
1964	<i>C_i</i>	C. R. Long	Tomich et al. (1968)
1972	<i>C</i>	<i>J. D. Dixon</i>	<i>Dixon (1973)</i>
May 1984	P	P. Sweet and J. Sweet	Sweet and Sweet (1985)
June–August 1985	P	J. Enbring et al.	Enbring et al. (1986)
1985	P	J. Swenson	Swenson (1986)
1986	P _i	A. Newman	Offshore Islet Project (2008)
1994	P _i	Anonymous	Offshore Islet Project (2008)
March–April 1989	P	G. C. Whittow et al.	Whittow et al. (1992)
1988–1989	P _i	G. C. Whittow et al.	Whittow et al. (1995)
February 2005	<i>C</i>	<i>K. Starr and F. Starr</i>	<i>Starr and Starr (2006)</i>
February 2008	P	H. Eijzena and D. Preston	Eijzena and Preston (2008)
October 2022–March 2023	<i>C</i>	<i>B. Najarian and K. Faccenda</i>	<i>This study</i>

Surveys varied in spatial extent, including comprehensive surveys (C) that covered most of the islet or partial (P) surveys that covered only part of the islet. Spatial extent was in some cases, inferred from the sources (C_i, P_i). The most robust survey results are used to characterize temporal vegetation dynamics and are italicized.

published secondhand (Bryan 1934, Tomich et al. 1968). Despite the variability in methods used for these surveys, they collectively offer a detailed history of the plant species occurring on Mānana Islet over the past 100 yr (Table 1).

New Vegetation Survey

To assess 40 yr of recovery following rabbit eradication, we conducted a new survey of Mānana during 4 visits to the islet between October 2022 and March 2023. Each survey typically started at 8:00 AM and concluded by 12:30 PM, and included myself and one other surveyor. The Department of Land and Natural Resources’ Division of Forestry and Wildlife (DOFAW) provided access to the islet, including boat transportation. Plant species were recorded along haphazard transects through the accessible portions of the islet, avoiding nesting seabirds and unpassable terrain. Binoculars and a telephoto lens camera were used to identify plants in inaccessible areas. All plant species were photographed, phenology was recorded (flowering or not), and position was taken using GPS. For species

that could not be identified in the field, samples were taken for lab identification.

Characterizing Vegetation Change across 100 Yr

Vegetation was characterized as species richness of native and non-native plants across the recent and historical surveys. Trends in species composition on Mānana are described, including new arrivals and species that were lost from the islet. Community turnover metrics are sensitive to missed species, giving rise to high potential for pseudo-turnover errors (Morrison 2016, Lisner and Lepš 2020). Because of the limited spatial extent of some of the earliest surveys (and resulting high likelihood of missed species), we take a conservative approach by not estimating community-turnover metrics and instead present broad patterns in plant species composition.

RESULTS

Over 96 yr, 21 surveys were performed on Mānana Islet, demonstrating the historical interest in the ecology of the islet (Table 1). In total, 60 species from 25 plant families were recorded during these surveys (Table 2). From

TABLE 2
Species Detected by Vegetation Surveys on Mānana Islet

Species	Plant Family	Life-Form	Survey Dates						
			1930	1936	1955	1964	1973	2006	2023
NATIVE									
<i>Argemone glauca</i>	Papaveraceae	He	X			X	X	X	X
<i>Boerhavia repens</i>	Nyctaginaceae	He/Th						X	X
<i>Chenopodium oahuense</i>	Amaranthaceae	He						X	X
<i>Eragrostis variabilis</i>	Poaceae	He/Gr						X	
<i>Heliotropium curassavicum</i>	Boraginaceae	He		X				X	
<i>Heteropogon contortus</i>	Poaceae	He/Gr	X	X	X				
<i>Ipomoea pes-caprae</i>	Convolvulaceae	He						X	X
<i>Panicum torridum</i>	Poaceae	He/Gr	X	X					
<i>Portulaca lutea</i>	Portulacaceae	He					X		
<i>Scaevola taccada</i>	Goodeniaceae	Ph						X	X
<i>Sesbania tomentosa</i>	Fabaceae	Ph							X
<i>Sesuvium portulacastrum</i>	Aizoaceae	He						X	X

(table continues)

TABLE 2 (Continued)

Species	Plant Family	Life-Form	Survey Dates						
			1930	1936	1955	1964	1973	2006	2023
<i>Sida fallax</i>	Malvaceae	He							X
<i>Solanum americanum</i>	Solanaceae	Th						X	X
<i>Tribulus cistoides</i>	Zygophyllaceae	Ph/Th						X	
<i>Waltheria indica</i>	Malvaceae	Ph	X	X	X			X	X
NON-NATIVE									
<i>Achyranthes aspera</i>	Amaranthaceae	He						X	X
<i>Ageratum conyzoides</i>	Asteraceae	Th	X	X		X		X	X
<i>Amaranthus spinosus</i>	Amaranthaceae	Th						X	X
<i>Amaranthus viridis</i>	Amaranthaceae	Th						X	X
<i>Atriplex semibaccata</i>	Amaranthaceae	Ph		X	X	X			
<i>Bidens alba</i>	Asteraceae	Th							X
<i>Boerhavia diffusa</i>	Nyctaginaceae	He	X		X	X	X	X	X
<i>Cenchrus ciliaris</i>	Poaceae	He/Gr						X	X
<i>Cenchrus echinatus</i>	Poaceae	He/Gr	X	X	X	X	X	X	X
<i>Chloris barbata</i>	Poaceae	He/Gr		X	X	X	X	X	X
<i>Coccinia grandis</i>	Cucurbitaceae	Ph						X	X
<i>Cocos nucifera</i>	Arecaceae	Ph			X	X	X	X	
<i>Cynodon dactylon</i>	Poaceae	He/Gr	X				X	X	X
<i>Cyperus javanicus</i>	Cyperaceae	He/Gr	X						
<i>Dactyloctenium aegyptium</i>	Poaceae	He/Gr			X	X	X	X	X
<i>Digitaria insularis</i>	Poaceae	He/Gr			X	X	X	X	X
<i>Distimake aegyptius</i>	Convolvulaceae	Th						X	X
<i>Eleusine indica</i>	Poaceae	He/Gr						X	X
<i>Emilia fosbergii</i>	Asteraceae	Th						X	X
<i>Eragrostis tenella</i>	Poaceae	He/Gr		X					
<i>Euphorbia hirta</i>	Euphorbiaceae	Th	X	X				X	X
<i>Lepidium didymum</i>	Brassicaceae	Th							X
<i>Leucaena leucocephala</i>	Fabaceae	Ph						X	
<i>Lysimachia arvensis</i>	Primulaceae	Th	X						
<i>Megathyrsus maximus</i>	Poaceae	He/Gr							X
<i>Mesosphaerum pectinatum</i>	Lamiaceae	He							X
<i>Neltuma pallida</i>	Fabaceae	Ph				X	X	X	
<i>Nephrolepis brownii</i>	Nephrolepidaceae	He							X
<i>Nicotiana tabacum</i>	Solanaceae	Th	X	X	X	X	X	X	X
<i>Oxalis corniculata</i>	Oxalidaceae	He							X
<i>Passiflora suberosa</i>	Passifloraceae	Ph						X	X
<i>Pluchea carolinensis</i>	Asteraceae	Ph				X		X	X
<i>Pluchea indica</i>	Asteraceae	Ph						X	X
<i>Portulaca oleracea</i>	Portulacaceae	Th	X	X	X	X	X	X	X
<i>Reichardia picroides</i>	Asteraceae	He						X	X
<i>Setaria adhaerens</i>	Poaceae	He/Gr		X	X	X	X	X	X

(table continues)

TABLE 2 (Continued)

Species	Plant Family	Life-Form	Survey Dates						
			1930	1936	1955	1964	1973	2006	2023
<i>Sida cordifolia</i>	Malvaceae	He				X			
<i>Solanum lycopersicum</i>	Solanaceae	Th	X		X	X	X		
<i>Sonchus oleraceus</i>	Asteraceae	Th	X			X	X	X	X
<i>Stapelia gigantea</i>	Apocynaceae	Su						X	X
<i>Tephrosia purpurea</i>	Fabaceae	Ch	X	X	X	X			
<i>Tetragonia tetragonoides</i>	Aizoaceae	He						X	
<i>Tridax procumbens</i>	Asteraceae	He						X	X
<i>Verbesina encelioides</i>	Asteraceae	Th						X	X
Total number of native species			4	4	2	1	2	11	10
Total number of non-native species			12	10	12	17	13	31	33
Total number of species			16	14	14	18	15	42	43

Native (including indigenous) species are listed first, followed by non-native species. Species presence is denoted with “X,” and absence is represented by empty cells. Life-forms were classified using the Raunkiaer system, where Ch-chamaephyte, Graminoid, He-hemicryptophyte, Ph-phanerophyte, Su-succulent, and Th-therophyte.

1930 to 1973, total species richness remained relatively constant, averaging 15.6 species observed in a given survey (Figure 2). In a comprehensive survey conducted in 2006, nearly 20 yr after the last rabbit was observed on Mānana, species richness increased to 42 as the vegetation began to recover from high levels of rabbit disturbance. Our 2022–2023 surveys recorded 43 species, representing a 176%

increase in diversity in the 40 yr following rabbit eradication (Figure 2). Native and non-native species generally followed similar patterns through the nearly 100 yr of surveys. However, non-native species outnumbered native species across all years, and the magnitude of difference is increasing over time (Figure 2). In the most recent survey in 2023, we detected 43 species on Mānana

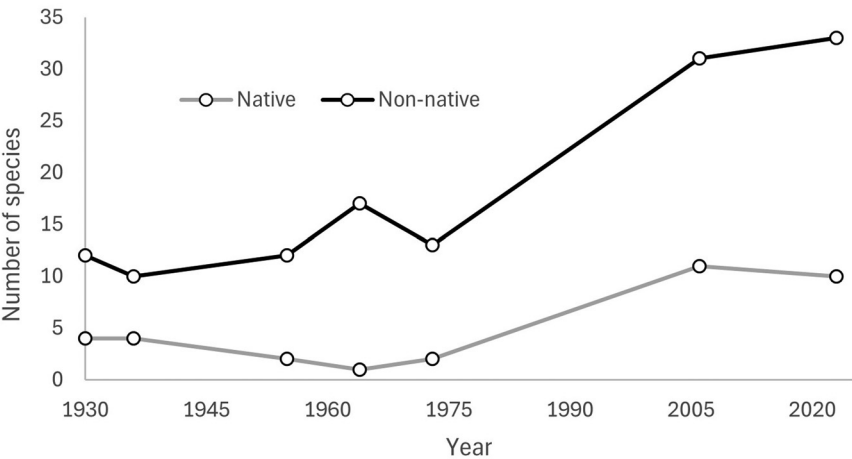


FIGURE 2. Patterns of species richness for native (grey line) and non-native (black line) across 100 yr of vegetation surveys on Mānana Islet.

(Table 2 and Supplementary Table 1). Of the 43 recorded species, 10 are native (endemic or indigenous), including a new record of *Sesbania tomentosa* occurring on the south-western ridge of the islet. Our surveys also recorded seven newly naturalized species to the islet, with six of those species categorized as non-native species (Table 2). These seven species likely dispersed from a source population on O'ahu, where they are present and naturalized. This increase in non-native plant species richness continues a trend that coincides with the rabbit eradication program. From 1930 to 1985, when rabbits were present on the islet, the average number of non-native species was 13, and the average number of native species was 3. In 1964, nearing the end of the rabbit eradication program, the number of invasive species rose to 17, while the number of native species dropped to 1. In 2006, the number of invasive species dramatically increased to 31, reaching 33 by 2023. Meanwhile, native species increased to 11 in 2006 but decreased to 10 in 2023. The trend of increasing species diversity was largely driven by naturalizations of non-native species on Mānana.

At present, the most abundant species is the non-native buffelgrass (*Cenchrus ciliaris*) which forms a near monoculture across much of the interior of the islet, in some areas this monoculture was interrupted by large stands of *Verbesina encelioides*. Other significant species include *Portulaca oleracea*, *Eleusine indica*, *Nicotiana tabacum*, *Setaria adhaerens*, *Sonchus oleraceus*, *Amaranthus viridis*, *A. spinosus*, and *Achyranthes aspera*, which often grow intermixed with buffelgrass. Mānana has a small beach on its South side dominated by naupaka (*Scaevola taccada*), pōhuehue (*Ipomoea pes-caprae*), *Cenchrus ciliaris*, *Eleusine indica*, several *Distimake aegyptius* and a single 'ohai (*Sesbania tomentosa*). On the two ridges leading to the high point of Mānana, we found *Coccinea grandis*, *Distimake aegyptius*, *Tridax procumbens*, *Emilia fosbergii*, *Stapelia gigantea*, *Bidens alba*, *Lepidium didymum*, *Verbesina encelioides*, *Reichardia picroides*, and a single *Nephrolepis brownii*. The partially collapsed cinder cone leading to the east-facing coastal bench was dominated by *Cenchrus echinatus* and *Scaevola taccada* interspersed with *Oxalis corniculata*, *Emilia fosbergii*,

Nicotiana tabacum, *Verbesina encelioides*, *Sonchus oleraceus*, *Dactyloctenium aegyptium*, and *Sesuvium portulacastrum*. The west-facing sea cliffs consisted mostly of *Cenchrus echinatus* with several very healthy and flowering pua kala (*Argemone glauca*).

DISCUSSION

Plant diversity on Mānana Islet has varied tremendously across the past 100 yr. While some of this temporal variability may reflect the dynamics of small populations near main islands, as predicted by island biogeography (MacArthur and Wilson 1967), surely most of this variability reflects the dominating effects of rabbits on the vegetation. When rabbits were abundant on Mānana, plant species richness was reduced to an average of 16 species, but then during the nearly 40 yr of recovery since rabbit eradication, the richness has increased by nearly threefold to 43 species in total. Unfortunately, this recovery in diversity has not resulted in a thriving native plant community, as non-native species far outnumber native species. The introduction of invasive plants is certainly a threat to Hawai'i's offshore islet communities, emphasizing the need for continued vegetation monitoring and management of these critical habitats.

Other islands with a high abundance of rabbits, such as Mauritius and Macquarie Island, have also shown increased plant diversity following the eradication of rabbits (North et al. 1994, Copson and Whinam 1998). However, studies focused on long-term vegetation recovery on islands and islets are rare, making it difficult to understand and predict trends across study sites (Kappes et al. 2021). Our surveys continue the temporal study of vegetation recovery post-mammalian extirpation that covers nearly 100 yr, leading to a better understanding of islet vegetation dynamics under heavy predation and long-term recovery after mammal removal.

On Mānana, the ratio of non-native to native species is over 3:1, and is significantly higher than the average of 1:1 found globally across islands (Sax and Gaines 2008). It places Mānana into the 26% of islands worldwide that have more non-native than native species

(Essl et al. 2019). Between 2006 and 2023, more than 20 yr after the last rabbit was observed on Mānana, seven new plant species established, six of which are non-native. The newly naturalized, non-native species also appear to displace other non-natives, which was seen within the crater on Mānana which was dominated by *Verbesina encelioides* in 2006 but is now almost entirely composed of *Cenchrus echinatus* (Figure 3). While there are no records of fires occurring on Mānana, this transition from an annual herb to a fire-prone annual grass raises the fuel-load while simultaneously increasing the intensity and frequency of fires (Weber 2017). However, the chance of ignition is low due to reduced anthropogenic activity. Similarly, research on Subantarctic Macquarie Island revealed that the number of introduced plant species generally increased

following the reduction of the European rabbit population (Copson and Whinam 1998).

For native species, recovery after the removal of rabbits is clearly not guaranteed, and in addition to the introduction of new non-native species, increases in the abundance of existing non-native species have also been reported. The eradication of rabbits on the Kerguelen Islands rapidly expanded the range of *Taraxacum officinale*. In addition, the diversity of native plant communities remained low in areas where rabbits were removed, compared to islands that had never experienced rabbit grazing (Chapuis et al. 2004). On Round Island in Mauritius, the non-native grass species *Chloris barbata* rapidly increased in abundance after rabbit eradication due to the release in grazing pressures, increase in available habitat, and high dispersal potential (North et al. 1994). In the same study, the native morning glory *Ipomoea pes-caprae*, also demonstrated more modest increases in abundance post-rabbits. However, the recovery of native species can be quite variable, depending on the species, duration of the invasion, and site characteristics (Zavaleta et al. 2001). For example, on Santa Cruz Island in California, native woody vegetation communities significantly recovered over a 28-yr period after the removal of feral sheep, without intensive conservation efforts (Beltran et al. 2014). Similarly, on an islet near Ibiza, the endangered *Medicago citrina* was able to successfully recover over a 5-yr period after the removal of herbivores (Rita et al. 2022). Yet, the appearance of an invasive insect now threatens this species, underscoring the complex challenges faced by native island species. On the Kermadec Islands, situated northeast of New Zealand, native species, including rare endemics, became more abundant following the removal of feral goats (Merlin and Juvik 1992, Vidal et al. 2000). The species *Veronica breviflora*, endemic to Raoul Island, vanished for almost 75 yr, reemerging only after goats were eradicated from the island.

The extirpation of European rabbits may have aided the rapid naturalization of non-native and invasive species on Mānana in the late 1970s. The removal of a dominant herbivore may have unintentionally triggered a

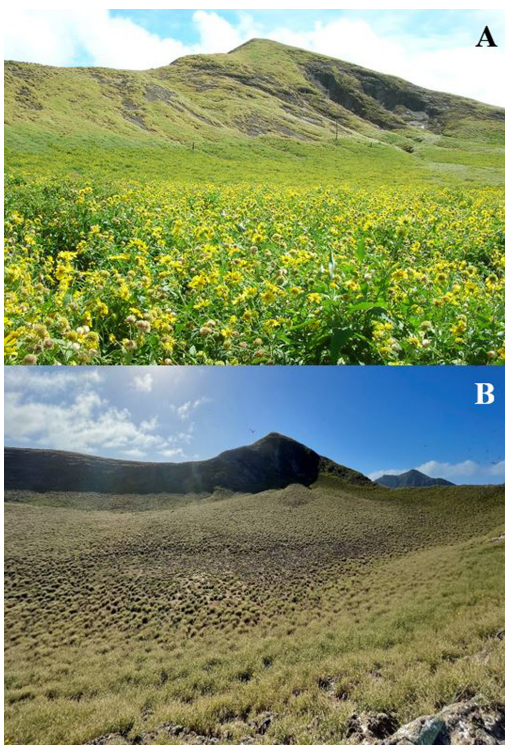


FIGURE 3. Vegetation within the interior of Mānana's western crater, which was predominantly covered by *Verbesina encelioides* in 2006 (A; courtesy of Forest and Kim Starr), has shifted by 2023 (B) to being dominated by *Cenchrus echinatus*.

top-down trophic cascade that facilitated the establishment and spread of non-native species. This shift allowed newly naturalized non-native species to complete their lifecycle and reduced grazing pressure on already naturalized non-native species. The denuding of Mānana due to the rabbits was well documented (Rabbit Island 1901, Swenson 1986), and vegetation change after the removal of rabbits has been studied on other Hawaiian Islands. On Lehua Island, which is approximately double the size of Mānana, there was a 53% increase in vegetation cover and a 71% increase in species richness after eradicating European rabbits, which were present on the island for at least 70 yr. However, the increase in cover was composed primarily of non-native species (112.8%) compared to natives (33.9%) (Eijzen 2011). Interestingly, newly naturalized species recorded on Lehua, including *Bidens pilosa* and *Boerhavia coccinea*, were conspecifics of new species recorded during our surveys, indicating a potential relationship between *O. cuniculus* and the suppression of these species. Rabbits can play a positive role in ecosystem function, enhancing species richness and soil fertility through bioturbation when they burrow. Their digging promotes soil turnover and moisture retention by reducing water runoff and capturing organic matter, including seeds (Fleming et al. 2014). However, frequent soil disturbance can increase turnover intensity and disturbance frequency, making islets more vulnerable to invasive species (Orbán et al. 2021). Ground-nesting seabirds similarly disturb the soil by digging nests, creating disturbed habitats that may favor the establishment of non-native plants (Merlin and Juvik 1992, Vidal et al. 2000).

The increase in non-native and invasive species on Mānana could also be attributed to land use changes and development in the nearby ahupua'a of Waimānalo, which transitioned from traditional agricultural production in the late nineteenth century to sugar production and tourism, as well as having large areas of land developed for military infrastructure in the early twentieth century. The development and industrialization of areas near Mānana have increased the number of

propagules from non-native species that can disperse to and potentially naturalize on the islet. The seven newly naturalized species documented in our surveys employ a range of dispersal mechanisms, including zoochory, anemochory, and hydrochory, with some utilizing multiple methods. For example, *Bidens alba*, which is widely established on O'ahu, disperses through both anemochory and epizoochory (Ramírez et al. 2012). Another potential dispersal vector is the non-native or indigenous birds that frequent the islet. During our January survey, we observed a group of approximately 15 Auku'u (*Nycticorax nycticorax*) on Mānana's east-facing tidal bench. These birds typically inhabit diverse aquatic environments, including suburban and urban waterways as well as aquaculture farms and may contribute to the dispersal of non-native epizoochorous species to the islet (HDLNR 2015). Anemochory and zoochory are the most probable dispersal mechanisms to Mānana, as tidal patterns do not facilitate hydrochorous transport from O'ahu, and the likelihood of anthropochory is low due to restricted access resulting from Mānana's designation as a sea-bird sanctuary. Brock and Daehler (2021) examined the link between socioeconomic development and plant naturalization trends in Hawai'i, finding that these trends closely follow shifts in economic drivers. In the same study, researchers found that naturalized species spread across the Main Hawaiian Islands at an average rate of 1.86 islands per decade. Over the 17 yr between our surveys and those conducted by Kim and Forrest Starr, seven new plant species dispersed and became naturalized on Mānana. This high number of naturalizations highlights the vulnerability of islets to the spread of non-native species from the Main Hawaiian Islands.

Managing islets requires a complex, multifaceted approach. According to Courchamp et al. (2003), effective management relies on pre-planning, comprehensive pre-control studies, careful method selection, and adequate political, financial, and public support. These guidelines are especially critical for offshore islet restoration where access is limited. For example, offshore islet monitoring programs could be expanded to assess non-native species using

the Hawai'i Pacific Weed Risk Assessment, which could inform managers on which species to prioritize for removal. Restoration plans for offshore islets also often aim to increase seabird nesting areas for future relocations. This could be achieved by removing invasive plant overgrowth, replanting native species, and actively monitoring native plants, including seed collection and targeted dispersal to promote species that tolerate local climatic conditions rather than those grown in greenhouses. Some researchers have also suggested using islets as refuges for critically endangered species (Carlquist 1989). However, given the challenges of habitat extremes, this idea may require careful evaluation on a case-by-case basis. On Mānana, several endemic species, such as 'ohai (*Sesbania tomentosa*), and pua kala (*Argemone glauca*), have been recorded on site, as shown in Figure 4. 'Ohai is a federally endangered species and currently ranked as endangered wherever found by the U.S. Fish and Wildlife Service (Department of the Interior 2024). The individual potentially outplanted 'ohai has increased in size, while at least a dozen individuals of pua kala can be found across the islet, including the windswept sea cliffs and large western crater. The persistence of 'ohai and pua kala provide some evidence that islets can act as refugia for imperiled endemic species adapted to these climates and warrants future study. Finally, understanding the potential additive



FIGURE 4. Pua kala (*Argemone glauca*) within the western crater of Mānana.

impacts of herbivore removal and climate change on non-native and native species is vital for successful long-term restoration plans (Kappes et al. 2021).

Globally, vegetation on islets is highly dynamic. However, many islets lack records of invasive species introduction and abundance through time (Walentowitz et al. 2023), making it difficult to understand whether these changes are due to current disturbances, disturbance legacy effects, or if the system is in the process of reaching an equilibrium state. Future research incorporating similar longitudinal studies on islets with reduced human disturbance could inform whether islets are intrinsically dynamic relative to their size and distance to source populations, as predicted by the theory of island biogeography, or if the changing species composition observed in disturbed habitats is due to invasion. Studies that also include species abundance, rather than presence/absence data, would also provide a more comprehensive picture of the level of invasion on islets. Finally, research focused on the effects of various means disturbance factors such as seabirds, projected abiotic alterations under climate change, and species abundance are needed to understand the drivers of vegetation dynamics on islets.

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